

Week 3

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + z = 5 \\ x + 3y + 0z = 1 \end{cases}$$

m - rows

$$\begin{cases} a_{11} & a_{12} & a_{13} & \dots & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & b_m \end{cases}$$

Augmented matrix

number of equations

Ex

$$n = 3 - \text{rows} \left\{ \begin{array}{cccc} \textcircled{1} & 2 & -3 & 4 \\ 3 & -1 & 1 & 5 \\ 1 & 3 & 0 & 1 \end{array} \right.$$

$(n+1) = (3+1) = 4 \text{ columns}$

augmented matrix

↳ play with augmented matrix

↳ get different matrix

↳ a different S.L.E

will have some solution to initial $\Rightarrow$ But much simpler

Aug $\Rightarrow$ S.L.E

$$\begin{bmatrix} 2 & 1 & 5 \\ 6 & 7 & 9 \end{bmatrix} \xrightarrow{?}$$

of rows $m = 2$

of columns $n+1 = 3$

$$\Rightarrow n = 2$$

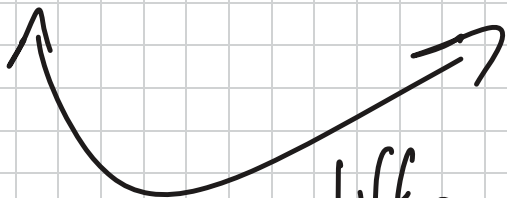
(x_1, x_2) $\xleftarrow{\quad}$ # of variables

$$2x_1 + 1x_2 = 5$$

$$6x_1 + 7x_2 = 9$$

$$\begin{cases} 1 & 2 & -3 & 4 \\ 3 & -1 & 1 & 5 \\ 1 & 3 & 0 & 1 \end{cases}$$

$$\begin{cases} 1 & 3 & 0 & 1 \\ 3 & -1 & 1 & 5 \\ 1 & 2 & -3 & 4 \end{cases}$$



different SLE
 same solution
 "elementary operations"

$$3 \quad -1 \quad 1 \quad 5 \quad \Rightarrow \quad (-3) [3 \quad -1 \quad 1 \quad 5]$$

Some solution

$$-9 \quad 3 \quad -3 \quad -15$$

$$\begin{array}{l} \text{LHS}_1 = \text{RHS}_1 \\ \text{LHS}_2 = \text{RHS}_2 \end{array} \quad \rightarrow \quad \text{LHS}_1 + \text{LHS}_2 = \text{RHS}_1 + \text{RHS}_2$$

$$\text{LHS}_1 + \# \text{LHS}_2 = \text{RHS}_1 + \# \text{RHS}_2$$

$$\text{LHS: } x + 2y - 3z + 3(3x - y + z)$$

$$= x + 2y - 3z + 9x - 3y + 3z$$

$$= 10x - y + 0 \cdot z$$

$$\text{RHS} = 19$$

Some solutions

$$\begin{Bmatrix} 2 & 1 & 5 \\ 6 & 7 & 9 \end{Bmatrix} \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \begin{Bmatrix} -10 & -13 & -13 \\ 6 & 7 & 9 \end{Bmatrix}$$

$$\begin{aligned} (2 \ 1 \ 5) + (-2) \cdot (6 \ 7 \ 9) &= \\ = (2 \ 1 \ 5) + (-12 \ -14 \ -18) &= \\ = (-10, -13, -13) \end{aligned}$$

*reating an owl to Clicks

REF matrix

First non zero # is always 1
("leading one")

rows of 0
must be at
the bottom

$$\begin{matrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{matrix} \left\{ \begin{array}{cccccc} 0 & 1 & 2 & 6 & 0 & 1 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right\}$$

Next row has to
Shift at least once

if all rules are met
it is in REF

RREF

↑
reduced

$$\begin{array}{c} C_1 \quad C_2 \quad C_3 \quad C_4 \quad C_5 \quad C_6 \\ \left[\begin{array}{cccccc} 0 & 1 & 2 & 6 & 9 & 1 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

or "0"

← matrix REF

BUT not RREF